

# S. Shayan Daneshvar

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📄 Scholar: scholar.google.ca/citations?user=NVHzLgOAAAAJ

## Education

- Jan 2023– March 2025 MSc, Computer Science, University of Manitoba, GPA: **4.5/4.5**, **Supervisor:** Dr. Shaowei Wang  
**Thesis Title:** Exploring Representation-level Augmentation and RAG-based Vulnerability Augmentation with LLMs for Vulnerability Detection  
**Courses:** Advanced Data Mining, Deep Learning with CNNs, Data-driven Software Engineering Research Methodologies, Image-based Generative Methods in Machine Learning (all **4.5/4.5 - A+**).
- Sep 2018– June 2022 BSc, Computer Engineering, K.N. Toosi University of Technology,  
GPA: **19.21/20** (US CGPA: **4/4, A+**), Ranked **1st** in class of **76**  
**Thesis Title:** Reflection Removal of In-vehicle Images (19.5/20) **Supervisor:** Dr. Nasihatkon  
**Selected Courses:** Computer Vision, Natural Language and Speech Processing, Database Design, Advanced Programming, Algorithm Design, Differential Equations, Compiler Design, Microprocessors & AVR Assembly, Discrete Mathematics, Hardware & Software Co-design, Numerical Methods, Software Engineering II, Internet Engineering. (all grades **20/20**)  
Software Engineering I (19.9/20), Linear Algebra (19.9/20), Operating Systems (19.75/20), Theory of Languages & Automata (19.75/20), Engineering Mathematics (19.5/20).

## Fields of Interests

- Generative AI, Large Language Models, Agentic AI, Deep Learning, Computer Vision & Image Processing
- Data Generation and Synthesis, Automated Software Engineering, AI4SE, SE4AI, MLOps

## Honors and Awards

- Feb 2025 **Faculty of Graduate Studies Research Completion Award**, C\$5000, University of Manitoba.
- Jan 2023– Jan 2025 **University of Manitoba Graduate Fellowship (UMGF)**, C\$28000, University of Manitoba.
- Jan 2023– Jan 2024 **International Graduate Student Entrance Scholarship (IGSES)**, C\$6500, University of Manitoba.
- Aug 2022 Straight MSc. in Artificial Intelligence **Admission Offer**, Sharif University of Technology.
- Sep 2019– July 2020 **Dean's List Inclusion**, Computer Engineering Faculty's Dean, K. N. Toosi University of Technology.
- Aug 2018 Ranked within the **Top 1%** Nationwide University Entrance Exam.

## Work Experience

- Nov 2025– present **LLM Research Engineer**, Huawei Canada – *Agentic Software Engineering*
- April 2025– September 2025 **Machine Learning Engineer**, MacDon Industries Ltd. – *Unsupervised Domain Adaptation (Contract via Zoomi Technologies)*
- Nov 2024– March 2025 **Machine Learning Engineer Intern**, MacDon Industries Ltd.  
*Generative AI for Digital Agriculture (Mitacs Accelerate)*
- May 2022– Dec 2022 **Senior Java Software Engineer**, Mahsan Co. – *Distributed Data-Analytics Platform*
- June 2022– Nov 2022 **Instructor**, Mahsan Co. – *Spring Cloud, Spring Framework, and Microservices Architecture*
- Nov 2021– May 2022 **Java Software Engineer**, Tosan Soha – *E-Wallet & Trading Platforms*
- May 2021– Nov 2021 **Java Backend Developer**, Pinket – *Chat & Shopping Services*
- Jan 2020– May 2021 **Java Instructor and Mentor (Part-time)**, Mapsa HR and Training
- June 2019– Sep 2019 **Intern**, Mapsa HR and Training – *Backend Web Development*

## Academic Experience

|                      |                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------|
| Jan 2023– March 2025 | Graduate Research Assistant, <b>ML4SE</b> , University of Manitoba, Supervisor: Dr. Shaowei Wang     |
| Sep 2023– Apr 2025   | Grader/TA, <b>Software Engineering II</b> , University of Manitoba, Instructor: Dr. Shaowei Wang     |
| Sep 2024– Dec 2024   | TA, <b>Applied Computational Intelligence</b> , University of Manitoba, Supervisor: Dr. Ahmed Ashraf |
| Jan 2024– Apr 2024   | Grader, <b>Machine Learning</b> , University of Manitoba, Instructor: Timothy Zapp                   |
| May 2023– Apr 2024   | Grader, <b>Software Engineering I</b> , University of Manitoba, Instructor: Robert Guderian          |
| Sep 2023– Dec 2023   | Grader, <b>Project Management</b> , University of Manitoba, Instructor: Dr. Shaowei Wang             |
| July 2021– July 2022 | Undergraduate Research Assistant, <b>Computer Vision</b> , KNTU, Supervisor: Dr. Behrooz Nasihatkon  |
| Sep 2021– Feb 2022   | Teaching Assistant, <b>Database Design</b> , KNTU, Instructor: Dr. Saeed Farzi                       |
| Feb 2021– July 2021  | Head Teaching Assistant, <b>Operating Systems</b> , KNTU, Instructor: Dr. Hamed Khanmirza            |
| Feb 2021– July 2021  | Head Teaching Assistant, <b>Algorithm Design</b> , KNTU, Instructor: Dr. Ali Ahmadi                  |
| Sep 2019– Aug 2020   | Head Teaching Assistant, <b>Advanced Programming</b> , KNTU, Instructor: Mehdi Zamanian              |

## Technical Skills

|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages         | Java, Python, C#, Kotlin, C/C++, Javascript & Typescript, Scala.                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Concepts          | OOP, GoF Design Patterns, Architectural Styles and Patterns, AOP, Prompt Engineering, RAG, SQL/NewSQL & NoSQL Databases, Concurrency and Parallelism, Event-Driven Programming, TDD/BDD, Web Development, Microservices Architecture, Android Development, Agentic AI Deep Learning, Generative Models, LLMs, Transformers, CNNs, RNN and LSTM, Computer Vision and Image Processing, AI for Software Engineering, Generative AI, Agile Software Development & Project Management (Scrum & Kanban) |
| ML Frameworks     | PyTorch, Tensorflow, Keras, Megatron-LM, HuggingFace, OpenCV, TensorBoard, vLLM.                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Web Frameworks    | Spring Framework(MVC, Webflux, Cloud, Security, Data, Boot), Angular 2+.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Tools             | LaTeX, Git, Gitlab, Docker, Nginx, Kafka, Hibernate, Flyway, Kubernetes.                                                                                                                                                                                                                                                                                                                                                                                                                           |
| IDEs              | IntelliJ IDEA, Pycharm, Visual Studio, VS Code, Claude Code, Codex.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Databases         | MySQL, Postgres, YugabyteDB, MS Server, MongoDB, Cassandra, SQLite.                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Other Libraries   | Hibernate, JUnit, Mockito, , JavaFX & WPF, Maven.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Operating Systems | Windows, Ubuntu Linux.                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Publications

Citations: 70+ - h-index: 3

**SS. Daneshvar**, Y. Nong, X. Yang, S. Wang, and H. Cai, "VulScribeR: Exploring RAG-based Vulnerability Augmentation with LLMs," TOSEM journal, 2026. [DOI](#). [Q1, IF: 7.0].

**SS. Daneshvar**, D. Tan, S. Wang, and C. Leung, "A Study on Mixup-Inspired Augmentation Methods for Software Vulnerability Detection," EASE Conference, 2025. [DOI](#). [CORE A].

N. Rezaei, **SS. Daneshvar**, B. Nasihatkon, S. Seidi, M. Rezazadeh, "The application of barcode readable assay and linear regression RGB analysis using a customized smartphone app in on-chip electromembrane extraction for simultaneous determination of heavy metal ions," Microchemical Journal, Volume 197, 2024. [DOI](#). Related to the **Visual Concentration Estimation** project. [Q1, IF: 5.4]

**SS. Daneshvar** and S. Wang, "GUI Element Detection Using SOTA YOLO Deep Learning Models," [preprint](#), 2024.

## Language

Farsi (Native), English (Advanced Proficiency - C1), Arabic (A2), French (A1)

Mar 2025– Mar 2027 **CELP-IP-General (English)**, Total: **11** | Reading: 11 | Listening: 11 | Speaking: 11 | Writing: 11

Sep 2021– Sep 2023 **TOEFL iBT (English)**, Total: **100** | Reading: 22 | Listening: 30 | Speaking: 22 | Writing: 26

## Selected Projects

### Important Coursework Projects

Feb - Apr 2024

#### Single Image Reflection Removal with MAMBA

Role: Researcher | Image-based Generative Methods in ML | Professor: Dr. Chris Henry  
Replicated the best SOTA Single Image Reflection Removal model (DSRNet), created an image-based version of a SOTA state-space model (Mamba/S6), and replaced attention-based modules of DSRNet with Mamba modules to improve the performance of the model. I also investigated the effect of the Cosine Annealing learning rate schedule and AdamW's weight decay.

Oct - Dec 2023

#### Exploring Vulnerability Augmentation with Mixup-based Representation-level Techniques

Role: Developer & Researcher | Advanced Data Mining | Professor: Dr. Carson K. Leung  
Used various SenMixUp-inspired methods to augment vulnerable code samples, after transforming them into a vector format and before training. I also developed a vulnerability-preserving heuristic to maintain the vulnerable state of code samples.

Mar - Apr 2023

#### Brain Tumor Segmentation and Survival Rate Prediction with 3DUNet variants

Role: Developer & Researcher | Deep Learning with CNNs | Professor: Dr. Ahmed Ashraf  
3D Segmentation of Brain Tumors on BraTS2020 dataset. I created, trained, and tested 3 variants of 3D UNet, namely Vanilla 3D UNet, Residual 3D UNet, and 3D UNet with a custom Attention mechanism. I also used the segmentation results to train an FCN to predict survival rates.

Jan - Mar 2023

#### GUI Element Detection using SOTA YOLO models

Role: Researcher | Data-driven Software Engineering | Professor: Dr. Shaowei Wang  
An empirical study on GUI Element Detection with the latest small-sized YOLO models (5, 6, 7, and 8). Specifically, I compared SOTA YOLO models for detecting Android GUI elements and showed that Text Views are often misclassified while Drawers and Switches are easier to detect.

June - July 2021

#### BSc Thesis, Reflection Removal of In-vehicle Images (with UNets)

Role: Developer & Researcher | BSc Thesis | Professor: Dr. Behrooz Nasihatkon  
I built and trained two variations of UNet to remove or decrease the effect of the reflections of in-vehicle images such that the resulting picture would look natural. For the dataset, I used a dataset of real-world streets and road images (CamVid) to synthesize a new dataset with reflections of various objects.

June - July 2021

#### Soccer Player Detection & Classification

Role: Developer | Computer Vision | Professor: Dr. Behrooz Nasihatkon  
Detection & Classification of players & referees from a recorded soccer match & visualization of their location on the soccer field. I used the KNN background subtraction algorithm with basic morphology techniques for the detection task. I labeled over 3K pictures for the classification task and trained a CNN. I also used perspective transform to merge 3 cameras' videos and map coordinates to a flat birds-eye view field.

Oct 2020–Feb 2021

#### Dapixi.ir

Role: Product Owner & Developer | Software Engineering | Professor: Dr. Mehdi Esnaashari  
Dapixi is a free photo sharing website similar to Pinterest. Responsibilities: back-end web development in Java, minor bug fixes in the Angular front-end, and development of the content-based & collaborative filtering photo recommender systems in Python.

Apr – July 2020

#### Database & Web Application Design of a Website

Role: Developer | Database Design | Professor: Dr. Saeed Farzi .  
Database Design & Fullstack web development of a website similar to divar.ir.

#### Other Major Projects

July 2021 – July 2022

#### Visual Concentration Estimation

Role: Android & Python Developer, Researcher | Supervisor: Dr. Behrooz Nasihatkon  
Detection of linear and QR barcodes including chemicals with CNNs, concentration estimation using linear and log-linear regressors, and outlier detection with PCA.

June – Sep 2021

#### Chibaladi.com

Role: Developer, Project Manager  
Backend design & development of the Chibaladi website. Chibaladi was a website for assessing developers' knowledge and recommending videos for closing their knowledge gaps.

## Academic Service

2026–present    Reviewer, **ACM Transactions on Software Engineering and Methodology (TOSEM)**  
 2026              Participant, **ICSE Shadow Program Committee (Shadow PC)**

## Certificates

August 2025              ACM Professional Member | ACM.  
 April 2025                Mitacs Accelerate | Mitacs.  
 September 2021        From Shallow to Deep Learning | ISMVIP.  
 August 2021             Fundamentals of Reinforcement Learning | IEEE KNTU Student Branch.  
 October 2019            Java Web Developer Certificate | Javacup Association.  
 Mar 2019 – present    Over 25 other certificates in various fields, mostly in Software Engineering | LinkedIn Learning.

## Hobbies

Playing Electric Guitar, Weightlifting, Running, Hiking, Playing Tennis, and Reading

## References

Dr. Shaowei Wang              Associate Professor, Department of Computer Science, University of Manitoba.  
                                          MSc supervisor and course instructor, Email: Shaowei.Wang@umanitoba.ca

Dr. Behrooz Nasihatkon       Assistant Professor, Department of Computer Engineering, K. N. Toosi University of Technology.  
                                          BSc supervisor, co-author, and course instructor, Email: nasihatkon@kntu.ac.ir

Dr. Christopher Henry        Associate Professor, Department of Computer Science, University of Manitoba.  
                                          Course instructor and Mitacs supervisor, Email: Christopher.Henry@umanitoba.ca

Dr. Haipeng Cai                Associate Professor, Department of Computer Science and Engineering, University at Buffalo.  
                                          Co-author, Email: haipengc@buffalo.edu

Dr. Carson Leung              Full Professor, Department of Computer Science, University of Manitoba.  
                                          Course instructor, co-author, and MSc thesis examiner, Email: Carson.Leung@umanitoba.ca

Dr. Shayan A. Tabrizi        Tech Lead, Software Architect and Manager, Mahsan Co., Tehran.  
                                          Supervisor and former colleague, Email: ShayanTabrizi@gmail.com

Dr. Ahmed Ashraf             Associate Professor, Department of Electrical and Computer Engineering, University of Manitoba.  
                                          Course instructor and TA supervisor, Email: Ahmed.Ashraf@umanitoba.ca

Dr. Babak Nasersharif        Associate Professor, Department of Computer Engineering, K. N. Toosi University of Technology.  
                                          BSc thesis examiner and course instructor, Email: bnasersharif@kntu.ac.ir

Dr. Shaiful Chowdury        Assistant Professor, Department of Computer Science, University of Manitoba.  
                                          MSc thesis examiner, Email: Shaiful.Chowdury@umanitoba.ca